

Beyond distribution shifts: Adaptive hyperspectral image classification at test time

Xia Yue^a, Anfeng Liu^b, Ning Chen^{c,*}, Chenjia Huang^d, Hui Liu^d, Zhou Huang^c, Leyuan Fang^{e,f}

^a School of Computer Science and Engineering, Central South University, Changsha, 410083, China

^b School of Electronic Information, Central South University, Changsha, 410083, China

^c Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing, 100871, China

^d School of Remote Sensing and Geomatics Engineering, Nanjing University of Information Science and Technology, Nanjing, 210044, China

^e College of Electrical and Information Engineering, Hunan University, Changsha, 410082, China

^f Peng Cheng Laboratory, Shenzhen, 518000, China

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ABSTRACT

Hyperspectral image (HSI) classification models are highly sensitive to distribution shifts caused by real-world degradations such as noise, blur, compression, and atmospheric effects. To address this challenge, we propose HyperTTA (Test-Time Adaptable Transformer for Hyperspectral Degradation), a unified framework that enhances model robustness under diverse degradation conditions. First, we construct a multi-degradation hyperspectral benchmark that systematically simulates nine representative degradations, enabling comprehensive evaluation of robust classification. Based on this benchmark, we develop a Spectral-Spatial Transformer Classifier (SSTC) with a multi-level receptive field mechanism and label smoothing regularization to capture multi-scale spatial context and improve generalization. Furthermore, we introduce a lightweight test-time adaptation strategy, the Confidence-aware Entropy-minimized LayerNorm Adapter (CELA), which dynamically updates only the affine parameters of LayerNorm layers by minimizing prediction entropy on high-confidence unlabeled target samples. This strategy ensures reliable adaptation without access to source data or target labels. Experiments on two benchmark datasets demonstrate that HyperTTA outperforms state-of-the-art baselines across a wide range of degradation scenarios. The code will be made publicly available at <https://github.com/halfcoder1/HyperTTA>.

1. Introduction

Hyperspectral remote sensing has attracted increasing attention in recent years due to its capability to acquire dense and continuous spectral information for each pixel across a wide range of wavelengths [1]. Such rich spectral signatures enable fine-grained discrimination of surface materials and have supported a wide spectrum of applications, including environmental monitoring [2], precision agriculture [3], mineral exploration [4], and urban planning [5]. Compared with conventional multispectral imagery, hyperspectral images (HSIs) provide significantly enhanced spectral resolution, allowing subtle material differences to be identified more reliably.

Despite these advantages, hyperspectral image classification remains a challenging task. The high dimensionality of HSI data often leads to the well-known curse of dimensionality [6], which, together with the scarcity of labeled samples [7], severely restricts model generalization.

Moreover, strong intra-class variability and high inter-class similarity [8–10] further complicate the learning process. In real-world scenarios, additional factors such as illumination variations, atmospheric effects, sensor noise, and acquisition conditions [11,12] introduce non-negligible distribution shifts between training and testing data, causing a significant performance degradation of models trained under ideal conditions. To address distribution shift without requiring access to labeled target data, test-time adaptation (TTA) has recently emerged as an effective and flexible learning paradigm [13,14]. Unlike conventional domain adaptation methods that rely on target-domain data during training [15–17], TTA adapts a pre-trained model directly at inference time using only unlabeled test samples. Typical TTA strategies include entropy minimization [18,19], normalization statistics adaptation, and self-ensembling mechanisms. This paradigm is particularly appealing for remote sensing applications, where test-time conditions are highly

* Corresponding author.

Email address: chenning0115@pku.edu.cn (N. Chen).

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dynamic and target annotations are often unavailable. A preliminary version of this work has been released as a preprint [20].

Although TTA has demonstrated promising results in natural image recognition and medical imaging tasks [21–25], its application to hyperspectral image classification remains largely underexplored. Most existing TTA methods are designed for low-dimensional RGB data and assume relatively simple input statistics, which are not directly applicable to the high-dimensional and spectrally correlated nature of HSIs [26]. Furthermore, traditional domain adaptation approaches often require multi-stage retraining procedures or access to labeled target data [27,28], limiting their scalability and practicality. In addition, there is currently a lack of benchmark datasets and standardized evaluation protocols for systematically assessing TTA methods under realistic hyperspectral degradation scenarios.

Motivated by these limitations, this paper proposes a novel TTA-oriented framework for robust hyperspectral image classification under diverse degradation conditions. The main contributions of this work are summarized as follows:

- We construct and release the first hyperspectral TTA benchmark that systematically simulates nine representative real-world degradations, including fog, stripe noise, additive Gaussian noise, zero-mean Gaussian noise, Poisson noise, convolutional blur, salt-and-pepper noise, JPEG compression, and deadline noise. This benchmark enables standardized evaluation of robustness and adaptability under diverse degradation scenarios.
- We design an SSTC module that incorporates a multi-level receptive field mechanism to capture spatial features at different scales, together with a label smoothing strategy to alleviate overfitting and enhance robustness under noisy supervision.
- We propose a confidence-aware TTA strategy termed the CELA module, which performs source-free adaptation by updating only the affine parameters of LayerNorm layers during inference. The adaptation is guided by entropy minimization on high-confidence target samples, improving generalization to unseen target distributions without requiring source data or target labels.
- By integrating the SSTC and CELA modules, we develop a unified TTA framework named **HyperTTA** for hyperspectral image classification under severe degradations. Extensive experiments on the PU and WHLK datasets demonstrate that HyperTTA consistently outperforms existing classification and TTA baselines in terms of accuracy, robustness, and generalization across diverse corruption types.

2. Related work

2.1. Deep learning for HSI classification

Deep learning approaches have attracted increasing attention in hyperspectral remote sensing due to their ability to automatically extract representative spectral-spatial features from raw data [29,30]. Convolutional Neural Networks (CNNs) have been extended to handle high-dimensional inputs through variants such as 3D-CNNs, which jointly model spectral and spatial information [31]. To improve efficiency while maintaining accuracy, Tri-CNN [32] introduced a multi-scale 3D-CNN architecture with three-branch feature fusion, enabling better classification with reduced computational cost.

Recently, Transformer-based models have demonstrated superior performance in HSI classification by leveraging self-attention mechanisms to capture long-range dependencies and complex spectral correlations [33–35]. For example, [36] introduced a Transformer with implicit conditional positional encoding to handle variable-length inputs, while [37] proposed a dual-attention structure that captures both global spatial-spectral context and local spectral features. To better integrate multiscale spatial-spectral information, [38] applied random masking and token-aware pooling. Hybrid models that combine CNN and ViT, such as [39], have also been proposed to bridge local and

global feature modeling. These Transformer architectures have significantly advanced the field, demonstrating strong generalization across complex scenes [40,41].

Despite these advances, deep learning-based HSI classifiers typically require large amounts of labeled data [42], which are expensive and labor-intensive to obtain [43]. Moreover, models trained on one dataset (source domain) often suffer severe performance drops when applied to data with different distributions (target domain) [44]. These challenges underscore the need for more robust methods that can maintain high classification accuracy under label scarcity and distribution shifts—especially under diverse real-world degradations, which are seldom considered in most existing work.

2.2. Robust and adaptive learning in HSI classification

In real-world remote sensing applications, HSIs are often degraded by sensor noise, atmospheric interference, compression artifacts, and optical distortions [45]. To enhance robustness, previous works have explored noise injection during training and denoising pre-processing pipelines [7,46]. However, most approaches focus on isolated degradation types and lack unified benchmarks to evaluate robustness under complex multi-source corruptions [47]. To address this, we construct a comprehensive multi-degradation dataset including nine representative corruption types (e.g., Gaussian, Poisson, stripes, blur, fog), enabling systematic evaluation of classification and adaptation models under diverse degradation scenarios.

Beyond degradation robustness, Domain Adaptation (DA) techniques aim to mitigate distribution shifts between source and target domains [48,49]. Existing DA methods align spectral-spatial representations through adversarial learning [50], manifold alignment [51], or transfer learning [52]. Recent advances further incorporate attention mechanisms [53], adaptive sampling [54], and masked self-distillation [55] to enhance cross-domain generalization.

However, conventional DA methods require access to target-domain data during training [15,16], which is often impractical in hyperspectral remote sensing. This limitation motivates TTA approaches that adapt models dynamically to unseen target domains without relying on target labels or prior data, forming the foundation of our proposed framework.

2.3. TTA concepts and methods

TTA has emerged as a flexible solution to address distribution shifts during inference, without requiring labeled target domain data. [56,57]. Unlike conventional domain adaptation methods that require access to source and target data during training [15–17], TTA adapts the model online at test time, typically by updating normalization statistics [18], minimizing prediction uncertainty [58], or leveraging self-ensembling techniques [59]. TTA has shown strong performance in fields such as natural image recognition and medical imaging, offering a practical balance between adaptation capability and annotation efficiency.

In the context of remote sensing, however, TTA remains underexplored. Most existing efforts primarily recalibrate batch normalization statistics on optical remote sensing data [60], while little attention has been paid to hyperspectral data, which presents unique challenges such as high spectral dimensionality, strong reflectance variability, and domain shifts caused by atmospheric and temporal factors [61]. Moreover, the large volume of HSI data makes real-time adaptation computationally demanding. These factors collectively hinder the direct application of existing TTA methods to hyperspectral scenarios.

To address these limitations, this paper proposes a TTA framework tailored specifically for HSI classification. Unlike prior TTA approaches focused on natural or medical images, our method performs entropy minimization on unlabeled target data while updating only the affine parameters of LayerNorm layers. This design maintains model compactness while enabling dynamic adaptation to unseen degradation patterns. Coupled with our multi-degradation hyperspectral dataset, the proposed

method offers a practical and efficient solution for real-world remote sensing applications where source data may be inaccessible and target distributions are non-stationary.

3. Method

In this section, we introduce **HyperTTA** (Test-Time Adaptable Transformer for Hyperspectral Degradation), our proposed framework designed to achieve robust HSI classification under diverse and unknown degradations. As shown in Fig. 1, the overall framework consists of three main components:

- **Multi-Degradation Simulator (MDS):** To simulate realistic distribution shifts, we construct a target-domain dataset using nine representative degradation types (e.g., fog, stripe, Gaussian noise), each controlled by specific parameters. This serves as the unlabeled input domain for TTA.
- **Spectral-Spatial Transformer Classifier (SSTC) Module:** We design a spectral-spatial Transformer architecture that incorporates a Multi-level Receptive Field (MRF) mechanism to capture hierarchical spatial information and applies Label Smoothing Loss (LSL) to improve generalization under noisy conditions. This model is trained only on clean, labeled source data.
- **Confidence-aware Entropy-minimized LayerNorm Adapter (CELA) Module:** During inference, we update only the affine parameters (γ, β) of LayerNorm layers by minimizing the prediction entropy on high-confidence unlabeled target samples. This lightweight and selective strategy enables robust and dynamic

adaptation to unseen degradations, while avoiding noisy updates, without requiring access to source data or target labels.

Together, these components allow HyperTTA to maintain high classification performance across diverse operational scenarios, even in the presence of severe data degradation and domain shifts.

To facilitate the presentation of our method, we first introduce the basic notations and problem setting. Let $\mathcal{X} \in \mathbb{R}^{H \times W \times C}$ denote an HSI, where H , W , and C represent the height, width, and number of spectral bands, respectively. The corresponding label space is denoted as $\mathcal{Y} = 1, 2, \dots, K$, where K is the total number of semantic classes.

We define the problem under a domain adaptation or TTA setting, where the data distributions differ between the training and testing environments. Let the **source domain** be defined as

$$D_s = \{(\mathbf{x}_i^s, y_i)\}_{i=1}^{N_s}, \quad (1)$$

where $\mathbf{x}_i^s \in \mathbb{R}^{H \times W \times C}$ denotes the i -th clean hyperspectral sample in the source domain, $y_i \in \{1, 2, \dots, K\}$ is the corresponding ground-truth label, N_s is the total number of labeled source samples, and K is the number of semantic classes.

The **target domain** is defined as

$$D_t = \{\mathbf{x}_j^t\}_{j=1}^{N_t}, \quad (2)$$

where $\mathbf{x}_j^t$ denotes the j -th unlabeled sample from the target domain. These samples are typically degraded, and no corresponding labels y_j^t are available during testing or adaptation. N_t denotes the number of target samples.

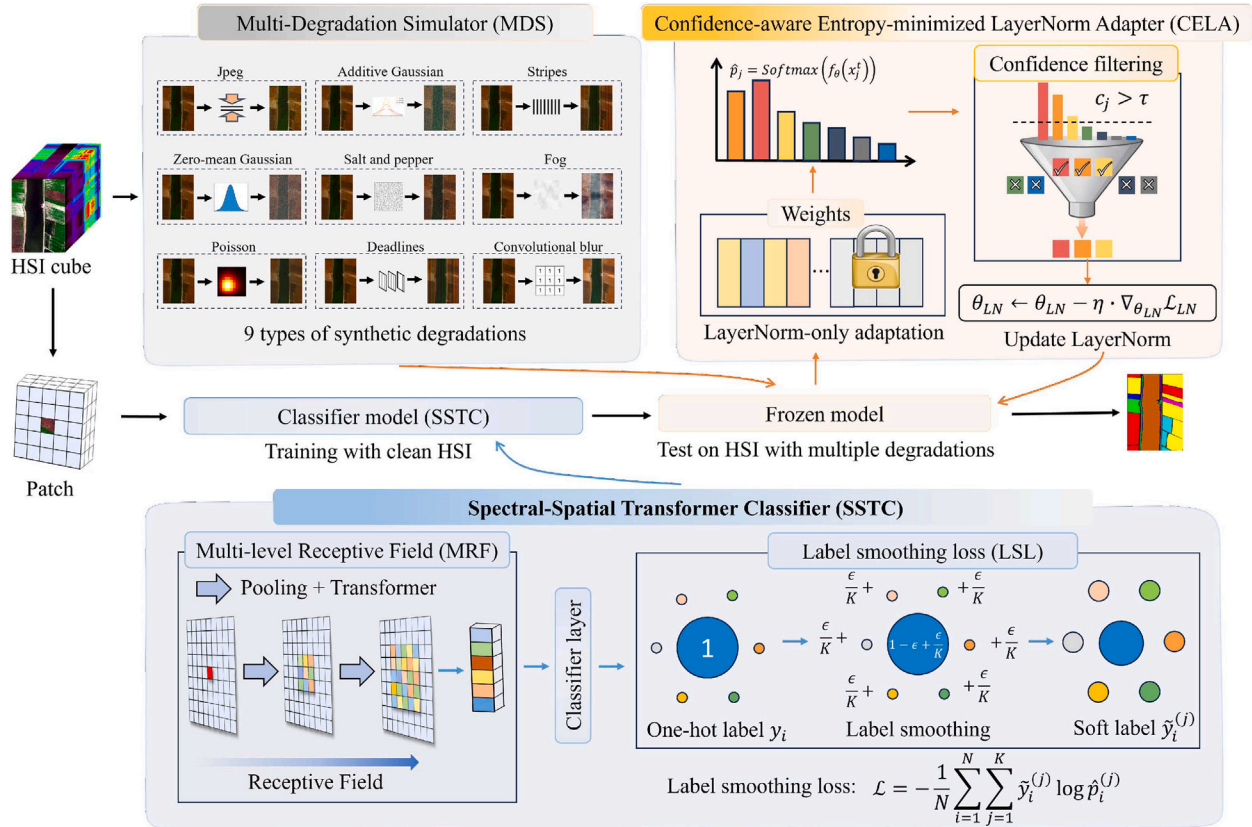


Fig. 1. Overview of the proposed HyperTTA framework. It consists of three main components: (1) a Multi-Degradation Simulator (MDS) that constructs unlabeled target-domain inputs by applying nine types of synthetic degradations to clean source data; (2) a Spectral-Spatial Transformer Classifier (SSTC) module trained on clean, labeled samples, incorporating a Multi-level Receptive Field (MRF) module and Label Smoothing Loss (LSL) module for robust representation learning; and (3) a Confidence-aware Entropy-minimized LayerNorm Adapter (CELA) module, which performs TTA by updating only the affine parameters of LayerNorm layers via entropy minimization on high-confidence target samples.

First, we construct a multi-degradation hyperspectral dataset that systematically simulates and organizes nine representative types of real-world degradations, including noise, blur, compression, and atmospheric effects. This dataset serves as a foundation for evaluating model robustness and supports the development of degradation-aware learning and adaptation techniques. Second, we aim to learn a model f_θ , parameterized by θ , that performs robust classification on the target domain under unknown and potentially degraded conditions. During TTA, the model parameters θ are updated based solely on the unlabeled target data, with the goal of minimizing prediction uncertainty or aligning feature distributions.

3.1. Multi-degradation simulator

To facilitate the study of robust HSI classification under complex and diverse real-world conditions, we construct a multi-degradation hyperspectral dataset that systematically simulates and organizes nine representative types of degradations as shown in Fig. 2. This dataset serves as a benchmark for evaluating noise-resilient models and TTA techniques.

The degraded sample is defined as

$$\tilde{\mathcal{X}} = \mathcal{N}_\phi(\mathcal{X}), \quad (3)$$

where $\mathcal{N}_\phi$ denotes a degradation operator parameterized by ϕ , and $\tilde{\mathcal{X}}$ is the resulting degraded hyperspectral image (HSI). Each operator simulates a specific real-world degradation, as described below.

- **JPEG Compression.** To simulate compression artifacts, JPEG compression is applied as:

$$\tilde{\mathcal{X}} = \text{JPEG}_q(\mathcal{X}), \quad (4)$$

where q denotes the compression ratio. In practice, JPEG compression is implemented using JPEG2000, which is widely adopted for hyperspectral imagery.

- **Zero-Mean Gaussian Noise (ZMGauss).** Each spectral band is first normalized to the unit interval:

$$\mathcal{X}_{\text{norm}}^{(c)} = \frac{\mathcal{X}^{(c)} - \min(\mathcal{X}^{(c)})}{\max(\mathcal{X}^{(c)}) - \min(\mathcal{X}^{(c)})}, \quad (5)$$

where $\mathcal{X}^{(c)} \in \mathbb{R}^{H \times W}$ denotes the c -th spectral band. Zero-mean Gaussian noise is then added:

$$\tilde{\mathcal{X}}^{(c)} = \text{clip}(\mathcal{X}_{\text{norm}}^{(c)} + \epsilon), \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad (6)$$

where σ^2 controls the noise variance and $\text{clip}(\cdot)$ enforces values in $[0, 1]$. After noise injection, the corrupted bands are rescaled

back to the original dynamic range to preserve the radiometric characteristics of each band.

- **Additive Gaussian Noise.** Following the same normalization in (5), a band-dependent noise level is sampled as:

$$\sigma_c \sim \mathcal{U}(0, \sigma_{\max}), \quad (7)$$

where $\sigma_{\max}$ denotes the maximum noise intensity. Gaussian noise is then generated as:

$$\epsilon_c \sim \mathcal{N}(0, \sigma_c^2), \quad (8)$$

and added to the normalized band:

$$\tilde{\mathcal{X}}^{(c)} = \text{clip}(\mathcal{X}_{\text{norm}}^{(c)} + \epsilon_c). \quad (9)$$

All normalized noisy bands are finally mapped back to the original value range.

- **Poisson Noise.** To model photon-counting noise, each spectral band is scaled as:

$$\mathcal{X}_{\text{scaled}}^{(c)} = \gamma_c \cdot \mathcal{X}^{(c)}, \quad (10)$$

where the scaling factor γ_c is computed from a target signal-to-noise ratio (SNR):

$$\gamma_c = \frac{\text{SNR}_{\text{lin}}}{\frac{1}{N} \sum_{i=1}^N \frac{(\mathcal{X}_i^{(c)})^2}{\mathcal{X}_i^{(c)} + \epsilon}}, \quad (11)$$

with $\text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/10}$, $N = H \times W$ denoting the number of pixels, and ϵ a small constant for numerical stability. Poisson noise is then applied as:

$$\tilde{\mathcal{X}}^{(c)} \sim \text{Poisson}(\mathcal{X}_{\text{scaled}}^{(c)}). \quad (12)$$

- **Salt-and-Pepper Noise.** Each normalized pixel (i, j) in band c is independently corrupted with probability p :

$$\tilde{\mathcal{X}}_{i,j}^{(c)} = \begin{cases} 0, & \text{with probability } \frac{p}{2}, \\ 1, & \text{with probability } \frac{p}{2}, \\ (\mathcal{X}_{i,j}^{(c)})_{\text{norm}}, & \text{with probability } 1 - p, \end{cases} \quad (13)$$

where p controls the corruption ratio.

- **Stripe Noise.** For each spectral band, the number of stripe columns is sampled as:

$$n_c \sim \mathcal{U}_{\text{int}}(a, b), \quad (14)$$

where $\mathcal{U}_{\text{int}}(\cdot)$ denotes a discrete uniform distribution. The selected columns are then overwritten by:

$$\tilde{\mathcal{X}}_{:,j}^{(c)} = \lambda_c, \quad \lambda_c \sim \mathcal{U}(0.6, 0.8), \quad (15)$$

where λ_c controls the stripe intensity.

- **Deadline Noise.** Deadline noise is simulated by randomly selecting several column locations and zeroing out consecutive columns to emulate sensor malfunction:

$$\tilde{\mathcal{X}}_{:,j:j+w_j-1}^{(c)} = 0, \quad w_j \in \{1, 2, 3\}, \quad (16)$$

where w_j denotes the width of each dead column. The number of dead columns in each spectral band is randomly sampled from a discrete uniform distribution:

$$n_c \sim \mathcal{U}_{\text{int}}(a', b'), \quad (17)$$

where (a', b') specifies the range of dead-column counts.

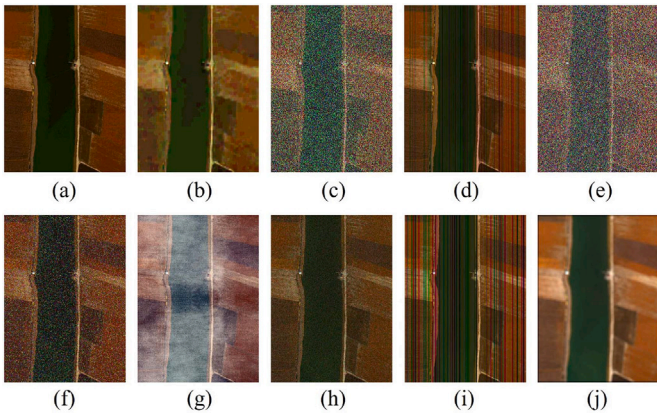


Fig. 2. False-color visualization of HSIs under multiple degradations: (a) Clean, (b) JPEG, (c) Additive Gaussian, (d) Stripes, (e) ZMGauss, (f) Salt & Pepper, (g) Fog, (h) Poisson, (i) Deadlines, (j) Convolutional Blur.

- **Convolutional Blur.** Convolutional blur is applied to simulate spatial resolution degradation caused by defocus or optical imperfections. Specifically, each spectral band is convolved with a spatial averaging kernel, optionally accompanied by mild additive noise to better approximate realistic imaging conditions:

$$\tilde{\mathcal{X}}^{(c)} = \mathcal{X}^{(c)} * K, \quad (18)$$

where $*$ denotes 2D convolution and

$$K = \frac{1}{k^2} \mathbf{1}_{k \times k}, \quad (19)$$

with k controlling the blur strength.

- **Fog Degradation.** To simulate atmospheric scattering effects in hazy scenes, we follow a wavelength-aware optical model that synthesizes fog on HSI inspired by [62]. For each spectral band c , the degraded image is generated as:

$$\tilde{\mathcal{X}}^{(c)} = \mathcal{X}^{(c)} \cdot t^{(c)} + A^{(c)} \cdot (1 - t^{(c)}), \quad (20)$$

where $\mathcal{X}^{(c)}$ is the clean input at band c , $A^{(c)}$ is the estimated global atmospheric light, and $t^{(c)}$ is the wavelength- and location-dependent transmission map. The transmission is modeled as:

$$t^{(c)}(x, y) = (t_1(x, y))^{\left(\frac{\lambda_0}{\lambda_c}\right)^{\gamma(x, y)}}, \quad (21)$$

with $t_1(x, y) = 1 - \omega \cdot \rho(x, y)$ being the base transmittance map derived from normalized fog density, and $\gamma(x, y)$ being an empirically regressed decay factor fitted over reference reflectance-transmittance pairs. Here, λ_c is the center wavelength of band c , and λ_0 is the shortest wavelength used as reference. This formulation introduces both spatial variation (via t_1 and γ) and spectral decay (via λ_c) to realistically model wavelength-dependent visibility in foggy conditions.

3.2. Spectral-spatial transformer framework with multi-level receptive field and label smoothing

We propose a spectral-spatial classification framework that avoids PCA-based band compression. Instead, it incorporates a multi-level receptive field mechanism and label smoothing regularization to improve generalization. The multi-level receptive field enables the model to capture both fine-grained local textures and broader contextual dependencies, which are crucial for handling diverse degradation patterns. Meanwhile, label smoothing mitigates overconfidence in predictions and improves model robustness under noisy or ambiguous inputs, thereby enhancing adaptation to multi-degradation scenarios.

3.2.1. Multi-level receptive field mechanism

To capture both fine-grained local textures and progressively larger spatial contexts, we design a spectral-spatial Transformer encoder with a hierarchical Multi-level Receptive Field (MRF). Given a hyperspectral sample $\mathbf{x}_i^s$, we extract a local patch $\mathbf{p}_i \in \mathbb{R}^{C \times w \times w}$ centered at the target pixel.

We first map the spectral dimension to a latent feature space using a lightweight convolutional stem:

$$\mathbf{X}_i = g(\mathbf{p}_i) \in \mathbb{R}^{d \times w \times w}, \quad (22)$$

where $g(\cdot)$ denotes a shared convolutional feature extractor implemented as a BN-Conv-BN-ReLU block, and d denotes the output channel dimension.

We flatten the spatial grid into a token sequence:

$$\mathbf{Z}_i^{(0)} = \text{Flatten}(\mathbf{X}_i) \in \mathbb{R}^{L \times d}, \quad L = w^2, \quad (23)$$

where $\text{Flatten}(\cdot)$ reshapes the spatial dimensions into a sequence of tokens while preserving the channel dimension. We then inject learnable

absolute positional embeddings:

$$\mathbf{Z}_i^{(0)} \leftarrow \mathbf{Z}_i^{(0)} + \alpha \mathbf{E}_{1:L}, \quad (24)$$

where $\mathbf{E} \in \mathbb{R}^{w^2 \times d}$ is a learnable embedding table and α is a learnable scaling factor.

Before multi-scale pooling, we apply a Transformer block on the full-resolution tokens to model local spectral-spatial interactions without changing the spatial resolution:

$$\mathbf{Z}_i^{(1)} = \text{Trans}^{(0)}(\mathbf{Z}_i^{(0)}), \quad (25)$$

where $\text{Trans}(\cdot)$ denotes a Transformer encoder block composed of multi-head self-attention and a position-wise feed-forward network with residual connections and LayerNorm. We then reshape $\mathbf{Z}_i^{(1)}$ back to a feature map $\mathbf{X}_i^{(1)} \in \mathbb{R}^{d \times w \times w}$ for hierarchical processing.

We progressively downsample the feature map using stage-wise max-pooling:

$$\mathbf{X}_i^{(t)} = \text{Pool}_{3,2,p_t}(\mathbf{X}_i^{(t-1)}), \quad t = 1, \dots, T, \quad (26)$$

where $\text{Pool}_{3,2,p_t}$ is a 3×3 max-pooling layer with stride 2 and $p_t \in \{0, 1\}$ is adaptively chosen to handle odd spatial sizes after downsampling. At each stage, the pooled feature map is flattened and processed by a Transformer block:

$$\mathbf{Z}_i^{(t)} = \text{Trans}^{(t)}(\text{Flatten}(\mathbf{X}_i^{(t)})). \quad (27)$$

This hierarchical design enlarges the effective receptive field across stages and yields multi-level representations.

At each stage, we extract the token (or equivalently the feature vector) corresponding to the center spatial position and form a list of center representations $\{\mathbf{z}_{i,\text{ctr}}^{(t)}\}_{t=1}^T$. In our implementation, we use the *final-stage* center representation for classification:

$$\mathbf{o}_i = h(\mathbf{z}_{i,\text{ctr}}^{(T)}) \in \mathbb{R}^K, \quad \hat{\mathbf{p}}_i = \text{Softmax}(\mathbf{o}_i), \quad (28)$$

where $h(\cdot)$ denotes a lightweight multi-layer perceptron (MLP) used as the classification head, $\mathbf{o}_i$ are the logits, and $\hat{\mathbf{p}}_i$ is the predicted class distribution. Specifically, $\mathbf{z}_{i,\text{ctr}}^{(t)}$ denotes the feature vector associated with the spatial center of the t -th stage feature map.

3.2.2. Label smoothing loss

Given the center token $\mathbf{z}_i^{(T)}$ produced by the MRF encoder, we obtain the classification logits via a linear classifier:

$$\mathbf{o}_i = \text{MLP}(\mathbf{z}_i^{(T)}) \in \mathbb{R}^K, \quad (29)$$

where $\text{MLP}(\cdot)$ denotes a lightweight multi-layer perceptron used as the classification head. The predicted class probability is obtained by applying the softmax function to the logits $\mathbf{o}_i = [o_i^{(1)}, \dots, o_i^{(K)}]^\top$:

$$\hat{p}_i^{(k)} = \frac{\exp(o_i^{(k)})}{\sum_{r=1}^K \exp(o_i^{(r)})}, \quad k = 1, \dots, K, \quad (30)$$

where $o_i^{(k)}$ denotes the k -th element of the logit vector $\mathbf{o}_i$, and $\hat{\mathbf{p}}_i = [\hat{p}_i^{(1)}, \dots, \hat{p}_i^{(K)}]^\top \in \mathbb{R}^K$ is the resulting predicted probability distribution.

To avoid overconfident predictions and improve generalization under degradations, we adopt label smoothing regularization [63]. During supervised training on the source domain, ground-truth labels are available. Let $y_i \in \{1, 2, \dots, K\}$ be the ground-truth label. We convert y_i into a one-hot vector $\mathbf{y}_i \in \{0, 1\}^K$, where $y_i^{(k)} = 1$ if $k = y_i$ and 0 otherwise. Label smoothing constructs a softened target distribution:

$$\tilde{\mathbf{y}}_i = (1 - \epsilon)\mathbf{y}_i + \epsilon \cdot \frac{1}{K}, \quad (31)$$

where $\epsilon \in [0, 1]$ is the smoothing factor. This softened supervision reduces the gradient dominance of the ground-truth class, alleviating

overconfident decision boundaries and improving calibration, which in turn enhances robustness when the model encounters noisy or degraded samples at test time. The training objective is then defined as the cross-entropy between the smoothed target and the predicted distribution:

$$\mathcal{L}_{LS} = -\frac{1}{N_s} \sum_{i=1}^{N_s} \sum_{k=1}^K \tilde{y}_i^{(k)} \log \hat{p}_i^{(k)}. \quad (32)$$

3.3. TTA with entropy minimization and confidence filtering

To improve the robustness of the model under distribution shift without access to source data, we adopt a TTA strategy based on entropy minimization [22], enhanced by a confidence-based sample selection mechanism. Specifically, we update only the affine parameters—namely, the scale γ and shift β —of the LayerNorm layers in the Transformer, while keeping all other model parameters frozen. This enables the model to dynamically adapt to target-domain statistics during inference, without requiring additional training or target labels.

Let $\mathbf{x}_j^t$ denote an unlabeled target-domain input sample, and let f_θ denote the pre-trained classification model parameterized by θ . During test time, we seek to update a subset of parameters $\theta_{LN} \subset \theta$ corresponding to $\{\gamma, \beta\}$ in each LayerNorm layer, such that the model minimizes prediction uncertainty. To quantify uncertainty, we use the entropy of the softmax output:

$$\mathcal{H}(f_\theta(\mathbf{x}_j^t)) = -\sum_{k=1}^K \hat{p}_j^{(k)} \log \hat{p}_j^{(k)}, \quad (33)$$

where $\hat{p}_j^{(k)}$ is the predicted probability for class k given the input $\mathbf{x}_j^t$.

To ensure stable and effective adaptation, we employ a hybrid confidence selection strategy. Let the confidence of sample $\mathbf{x}_j^t$ be defined as:

$$c_j = \max_k \hat{p}_j^{(k)}, \quad (34)$$

and define a confidence threshold $\tau \in [0, 1]$. Let $\mathcal{I}_\tau = \{j \mid c_j > \tau\}$ be the set of samples above the threshold. To avoid the case where too few samples are selected (i.e., $|\mathcal{I}_\tau|$ is small), we also define $\mathcal{I}_{\text{top}} \subset \{1, \dots, B\}$ as the indices corresponding to the top 30% most confident samples in the mini-batch. We then define the final selected index set as:

$$\mathcal{I} = \begin{cases} \mathcal{I}_\tau, & \text{if } |\mathcal{I}_\tau| \geq 0.3B \\ \mathcal{I}_{\text{top}}, & \text{otherwise} \end{cases} \quad (35)$$

The adaptation loss is computed only on the selected samples:

$$\mathcal{L}_{LN} = \frac{1}{|\mathcal{I}|} \sum_{j \in \mathcal{I}} \mathcal{H}(f_\theta(\mathbf{x}_j^t)). \quad (36)$$

During inference, for each mini-batch of target samples, we update only the LayerNorm parameters $\{\gamma, \beta\}$ by minimizing the entropy-based adaptation loss, while freezing all other model parameters. Specifically, the updates are:

$$\gamma \leftarrow \gamma - \eta \cdot \frac{\partial \mathcal{L}_{LN}}{\partial \gamma}, \quad \beta \leftarrow \beta - \eta \cdot \frac{\partial \mathcal{L}_{LN}}{\partial \beta}, \quad (37)$$

where η is the learning rate for test-time optimization.

We adopt an episodic adaptation strategy, in which the LayerNorm parameters $\{\gamma, \beta\}$ are reset to their original pre-trained values before adapting to each test sample. This design ensures that adaptation is performed independently for each test instance and effectively prevents parameter drift across different test episodes.

4. Experiments

4.1. Experimental settings

4.1.1. Datasets

To verify the effectiveness of our algorithm, we applied the algorithm to two public datasets: the Pavia University dataset (PU) and the WHU-Hi-Longkou dataset (WHLK):

- The Pavia University dataset as shown in Fig. 3, captured by the ROSIS sensor in 2003, this urban scene around the University of Pavia (Italy) contains 610×340 pixels with a 1.3 m spatial resolution and 115 spectral bands (430–860 nm). After removing 12 noisy bands, 103 bands remain. The dataset includes 9 land-cover classes.
- The WHU-Hi-Longkou dataset [64,65] as shown in Fig. 4. Acquired on July 17, 2018, in Longkou, Hubei, China, using a Headwall Nano-Hyperspec sensor mounted on a DJI M600 Pro UAV at 500 m altitude. The imagery (550×400 pixels) covers 270 bands (400–1000 nm) with a 0.463 m spatial resolution, depicting 9 agricultural land-cover types.

The detailed specifications of the two datasets are provided in Table 1.

4.1.2. Comparison method

To evaluate the effectiveness of the proposed method, we compare it with several representative baselines and state-of-the-art TTA techniques. SSRN [66] (Spectral-Spatial Residual Network) serves as a strong fully supervised baseline without any adaptation. SSFTT [67] is a Transformer-based model that incorporates spectral-spatial fusion and is trained under a fully supervised setting, serving as an advanced backbone for comparison. CASST [68] is a Transformer-based architecture that incorporates spatial and channel attention mechanisms to

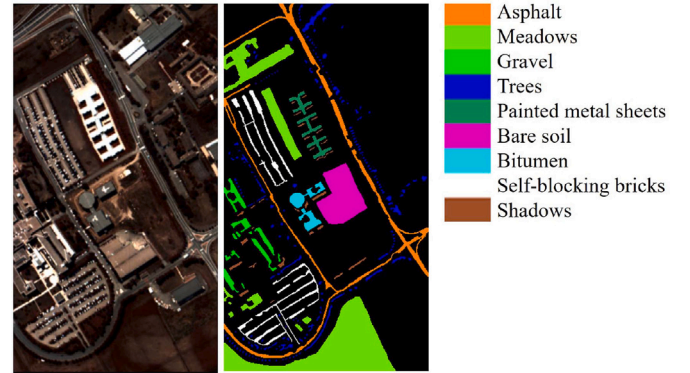


Fig. 3. Visualization of the Pavia University dataset: false-color composite image, ground-truth map, and class legend.

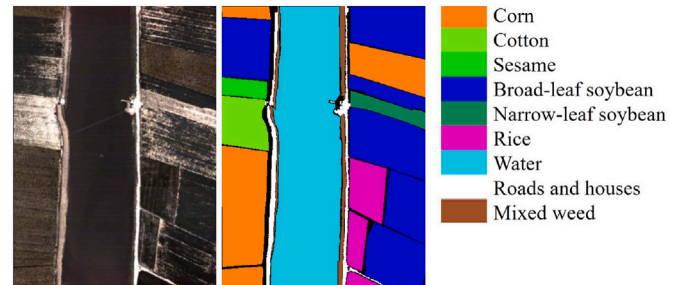


Fig. 4. Visualization of the WHU-Hi-Longkou dataset: false-color composite image, ground-truth map, and class legend.

Table 1
Detailed data of the PU dataset and the WHLK dataset.

Class	Pavia University		WHU-Hi-LongKou	
	Class Name	Samples	Class Name	Samples
1	Asphalt	6631	Corn	34,511
2	Meadows	18,649	Cotton	8374
3	Gravel	2099	Sesame	3031
4	Trees	3064	Broad-leaf soybean	63,212
5	PaintedMetalSheets	1345	Narrow-leaf soybean	4151
6	BareSoil	5029	Rice	11,854
7	Bitumen	1330	Water	67,056
8	SelfBlockingBricks	3682	Roads and houses	7124
9	Shadows	947	Mixed weed	5229
Total		42,776		204,542

enhance local feature modeling for HSI classification. MASSFormer [69] employs multi-scale attention to jointly model spectral and spatial features, achieving robust performance under various degradations. Tent [22] adapts the model during inference by minimizing the entropy of predictions on unlabeled target samples, without access to source data. SoTTA [70] improves robustness by introducing confidence-based filtering and entropy-sharpness minimization, and adapts only the parameters of LayerNorm during testing. CoTTA [71] performs test-time self-training with EMA modeling and stochastic restoration to enable continual adaptation under distribution shifts. SAR [72] uses structure-aware reconstruction to preserve spatial-spectral consistency for robust domain adaptation in HSI classification. CFA [73] enhances the robustness of Vision Transformers at test time by introducing a consistency-based filtering approach. BNAdapt [74] is a simple TTA method that updates Batch Normalization statistics using target data, enabling domain adaptation without requiring source labels.

4.1.3. Implementation details

All experiments were conducted on a workstation equipped with a 13th Gen Intel(R) Core(TM) i7-13700KF CPU (3.4 GHz, 16 cores), 64 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. The proposed model was implemented using PyTorch. During the initial training phase, the patch and batch sizes were set to 15 and 64 for the PU dataset, and 21 and 32 for the WHLK dataset, respectively. The training was conducted using the Adam optimizer with a learning rate of 0.001, for 20 epochs on the PU dataset and 10 epochs on the WHLK dataset. The smoothing factor $\epsilon = 0.05$. For supervised training, 20% of the clean samples and their corresponding labels from each class were used. Importantly, no degraded samples were involved during the training phase, ensuring that the model was trained solely on clean data without exposure to any degradation patterns. The remaining 80% were transformed into degraded inputs using our proposed multi-degradation simulation strategy, and their labels were discarded during adaptation. These degraded samples were used exclusively during the TTA phase as unlabeled target data, eliminating any risk of data leakage or noise memorization during training. These degraded samples were then used as TTA inputs. During the TTA phase, model parameters

were updated for a single epoch using the stochastic gradient descent (SGD) optimizer with a learning rate of 0.001 and a batch size of 64. To ensure the reliability of the adaptation process, only high-confidence target samples—selected based on a confidence threshold of $\tau = 0.8$ —were utilized for parameter updates. All experimental results are reported as the average over 3 independent runs to ensure statistical robustness. The specific degradation simulation settings for both the PU and WHLK datasets are summarized in Table 2.

4.2. Comparison of classification models under multi-degradation

To assess the robustness and generalization of the proposed SSTC model, we compare it with representative methods including SSRN [66], SSFTT [67], CASST [68], and MASSFormer [69]. All models are trained on clean data and tested under multiple simulated degradations on the PU and WHLK datasets to evaluate their resilience to quality deterioration.

As shown in Fig. 5, SSTC consistently outperforms the compared models on the PU dataset across most degradation types, particularly under severe conditions such as Poisson and stripe noise. Moreover, SSTC achieves the highest average OA, highlighting its strong robustness and stability.

Similarly, Fig. 6 shows that SSTC maintains superior and more stable performance on the WHLK dataset under various degradations, including ZM Gaussian and salt-and-pepper noise. Although its average OA is close to MASSFormer, SSTC achieves higher consistency across all corruption types.

Overall, SSTC demonstrates higher accuracy and stability than competing models on both datasets, confirming its robustness to distribution shifts and suitability as a foundation for subsequent TTA methods.

4.3. Comparison of TTA methods

To assess the effectiveness of the proposed HyperTTA framework, we compare it with several state-of-the-art TTA methods, including

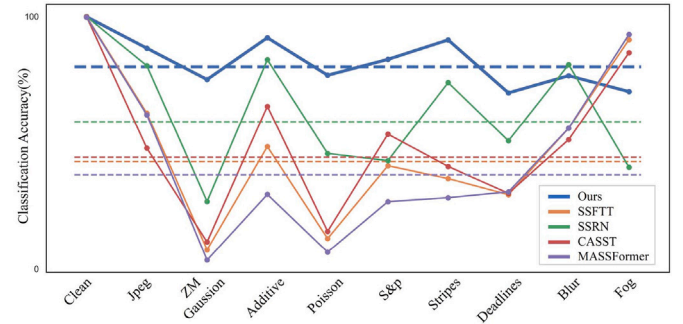


Fig. 5. Overall accuracy (OA) of different models under various degradation types on the PU dataset. Solid lines indicate the per-degradation performance, while dashed lines denote the average OA across all degradations.

Table 2
Parameters used to simulate nine types of degradations on the PU and WHLK datasets.

Degradation type	Parameter	Description	PU	WHLK
JPEG compression	q	Compression ratio for JPEG2000 encoding	10	15
Additive Gaussian noise	$\sigma_{\max}$	Maximum standard deviation of band-wise additive Gaussian noise	0.20	0.25
Salt-and-pepper noise	p	Noise density (probability of pixel corruption)	0.10	0.09
Convolutional blur	k	Kernel size of the spatial averaging filter	3	3
Zero-mean Gaussian noise	σ	Standard deviation of zero-mean Gaussian noise	0.25	0.15
Poisson noise	SNR_{dB}	Target signal-to-noise ratio in decibels	-10	15
Stripe noise	(a, b)	Range of integer-valued stripe counts per band	(30, 35)	(35, 40)
Deadline noise	(a', b')	Range of integer-valued dead-column counts per band	(30, 35)	(25, 30)
Fog degradation	ω	Fog density coefficient controlling atmospheric scattering strength	0.30	0.20

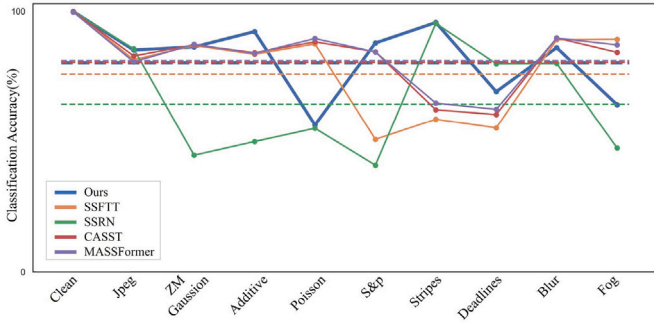


Fig. 6. Classification accuracy (OA) of different models under various degradation types on the WHLK dataset. Solid lines denote per-degradation performance, while dashed lines represent the average OA across all degradations.

Tent [22], CoTTA [71], BNAdapt [74], CFA [73], SAR [72], and SoTTA [70]. For a fair comparison, Tent, CoTTA, and BNAdapt are implemented using SSRN as the backbone, while CFA, SAR, and SoTTA adopt SSFTT as their base model. Our HyperTTA is built upon the proposed SSTC backbone. All methods are evaluated on the same degraded test sets of the PU and WHLK datasets.

Table 3 reports the classification performance in terms of Overall Accuracy (OA), Average Accuracy (AA), and Kappa coefficient (Kappa), where OA measures the overall proportion of correctly classified pixels, AA denotes the mean class-wise accuracy, and Kappa evaluates the agreement between predictions and ground truth while accounting for chance agreement. The visualized OA performance is shown in Fig. 7. Compared with all baselines, the proposed HyperTTA achieves higher classification accuracy across most degradation types. Specifically, it outperforms other methods under challenging conditions such as additive noise (89.15%), Poisson noise (81.52%), and deadlines (73.24%), while maintaining strong performance on milder corruptions such as JPEG and S&P noise. Moreover, HyperTTA achieves the highest average performance, with an OA of 82.48%, AA of 73.59%, and Kappa of 76.27%, demonstrating superior robustness and generalization capability. Fig. 8 further visualizes the classification maps produced by different TTA methods under various degradations. It can be observed that our method preserves finer spatial structures and generates

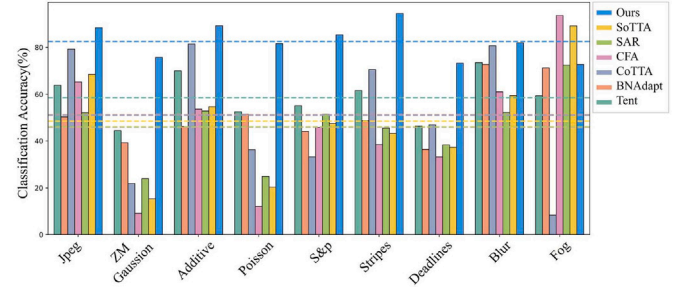


Fig. 7. Classification performance (OA) of different TTA methods under diverse degradation types on the PU dataset. Each group of bars corresponds to a specific degradation type, and the dashed lines represent the average accuracy of each method across all degradations.

fewer artifacts, particularly under heavy noise conditions (e.g., Poisson, stripes, ZMGauss), while other methods often produce fragmented or noisy predictions.

Table 4 shows the classification results of various TTA methods on the WHLK dataset under multi-degradations, and the visualized OA performance is shown in Fig. 10. Overall, our HyperTTA method outperforms all baselines by a significant margin in terms of OA, AA, and Kappa metrics. In particular, it achieves an average OA of 87.40%, AA of 86.66%, and Kappa of 84.68%, which are markedly higher than the second-best method, SoTTA (OA: 77.99%). HyperTTA maintains high performance across nearly all degradation types, especially on challenging conditions such as Zero-Mean Gaussian (97.08%), Salt & Pepper noise (97.25%), and fog (88.86%). The classification maps in Fig. 9 further illustrate the effectiveness of different TTA methods under various degradations. For spatially homogeneous classes (e.g., water bodies and expansive farmland), larger receptive fields help enforce spatial consistency and mitigate local perturbations caused by degradations, while smaller receptive fields preserve class boundaries. As a result, HyperTTA produces more spatially coherent classification maps with fewer fragmented regions compared with the competing methods. This improvement can be attributed to the complementary effects of the SSTC and CELA modules. Specifically, the MRF design in SSTC enhances spatial context modeling across different scales, which helps preserve

Table 3
Classification results on the PU dataset.

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt & Pepper	Stripe	Deadline	Blur	Fog	Avg.
Tent	OA (%)	63.83	44.39	69.97	52.59	55.15	61.63	46.33	73.50	59.39	58.53
	AA (%)	53.23	32.92	57.30	43.35	44.81	56.04	41.51	60.39	37.20	47.42
BNAdapt	Kappa $\times 100$	50.62	24.74	56.91	36.89	38.60	49.34	30.49	62.16	40.87	43.96
	OA (%)	50.27	39.22	46.13	51.46	44.09	48.60	36.43	72.73	71.22	51.35
	AA (%)	33.23	20.30	28.66	31.94	25.45	32.72	21.58	59.65	53.97	34.61
	Kappa $\times 100$	31.08	15.65	26.35	33.70	23.32	30.59	16.71	61.36	59.47	33.80
CoTTA	OA (%)	79.23	21.89	81.38	36.24	33.24	70.48	46.83	80.63	8.39	51.59
	AA (%)	78.94	44.23	81.85	59.60	56.03	82.39	59.29	74.61	19.89	61.65
	Kappa $\times 100$	73.03	15.64	75.34	29.41	26.55	63.77	37.23	72.82	4.78	44.29
	OA (%)	65.26	9.18	53.67	12.17	45.77	38.47	33.22	61.04	93.54	45.70
CFA	AA (%)	55.08	10.27	59.05	11.91	54.77	46.68	39.29	70.92	91.40	48.82
	Kappa $\times 100$	54.60	-5.71	45.09	-3.07	37.79	29.12	21.54	53.40	91.43	36.13
SAR	OA (%)	52.16	23.93	52.87	24.89	51.48	45.48	38.28	52.21	72.36	45.41
	AA (%)	42.58	12.55	42.61	11.23	41.93	37.79	31.47	48.72	69.81	37.52
	Kappa $\times 100$	34.57	-7.98	34.56	-5.45	32.81	25.35	18.77	35.90	62.25	25.31
SoTTA	OA (%)	68.44	15.35	54.71	20.40	47.39	43.25	37.31	59.45	89.12	48.16
	AA (%)	57.01	21.21	62.07	25.41	60.01	53.36	41.16	70.62	86.85	53.52
	Kappa $\times 100$	57.67	1.76	46.28	5.68	39.54	33.30	23.98	51.24	85.72	38.68
Ours	OA (%)	88.32	75.71	89.15	81.52	85.32	94.43	73.24	81.92	72.69	82.48
	AA (%)	81.13	60.67	83.83	67.01	76.42	89.98	71.11	77.30	54.85	73.59
	Kappa $\times 100$	84.07	67.07	85.15	75.01	80.06	92.63	65.71	74.58	62.17	76.27

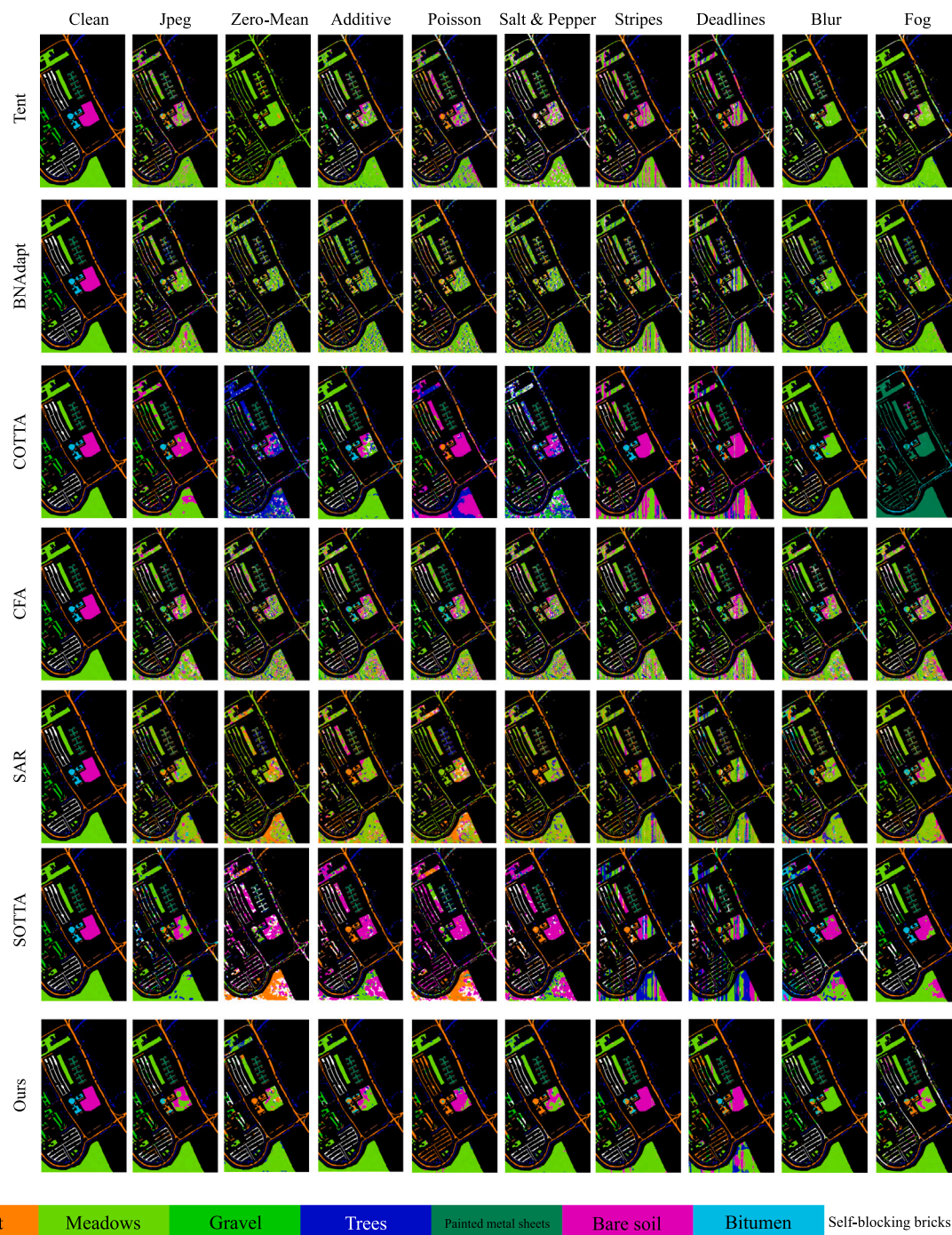


Fig. 8. Visual classification results on the PU dataset under different degradation types after applying different TTA method. Each row corresponds to a method, and each column represents a specific degradation.

Table 4
Classification results on the WHLK dataset.

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt & Pepper	Stripe	Deadline	Blur	Fog	Avg.
Tent	OA (%)	79.41	71.61	76.00	72.57	61.44	78.82	69.27	78.40	65.63	72.79
	AA (%)	56.32	54.80	57.65	51.00	49.32	60.10	48.45	58.04	48.31	53.89
	Kappa $\times 100$	72.37	62.09	67.79	63.29	48.35	72.06	59.51	71.22	54.14	63.43
BNAdapt	OA (%)	41.31	45.86	45.32	53.12	30.16	45.82	40.57	59.14	54.55	46.65
	AA (%)	26.37	27.93	28.37	37.61	20.41	30.64	25.28	41.46	39.97	31.78
	Kappa $\times 100$	24.05	27.77	27.10	37.70	7.35	29.65	22.81	45.66	39.74	29.87
CoTTA	OA (%)	72.17	37.98	42.45	47.62	19.49	93.41	72.20	68.13	45.02	55.83
	AA (%)	75.82	52.38	57.89	54.29	36.57	94.67	71.40	77.92	50.31	63.47
	Kappa $\times 100$	66.31	30.62	34.33	39.23	14.74	91.50	65.95	62.06	37.02	49.64
CFA	OA (%)	67.03	58.10	53.64	71.45	50.04	45.60	42.08	71.82	69.23	58.89
	AA (%)	49.28	44.00	38.62	62.81	41.32	33.03	25.59	67.18	63.83	47.52
	Kappa $\times 100$	59.17	50.05	45.15	64.88	41.60	33.03	29.91	65.48	62.60	50.65
SAR	OA (%)	53.19	55.90	53.34	55.10	47.33	44.61	43.27	56.29	57.15	51.80
	AA (%)	27.57	33.20	30.37	30.58	27.07	23.61	19.68	42.38	43.15	30.63
	Kappa $\times 100$	36.44	39.56	36.09	37.19	27.64	24.14	22.64	41.48	33.97	33.46
SOTTA	OA (%)	76.10	85.53	83.73	87.64	82.76	60.15	52.76	83.34	82.84	77.99
	AA (%)	42.44	52.34	49.89	57.20	45.57	29.06	24.87	72.01	69.23	49.51
	Kappa $\times 100$	68.22	80.37	77.98	83.37	76.70	45.30	38.13	78.42	77.98	69.72
Ours	OA(%)	87.72	97.08	98.70	60.58	97.25	99.10	65.16	92.16	87.86	87.40
	AA(%)	93.29	90.01	95.81	74.44	90.38	98.36	76.66	85.30	64.69	86.66
	Kappa $\times 100$	85.32	96.15	98.29	54.11	96.38	98.82	58.98	89.83	83.27	84.68

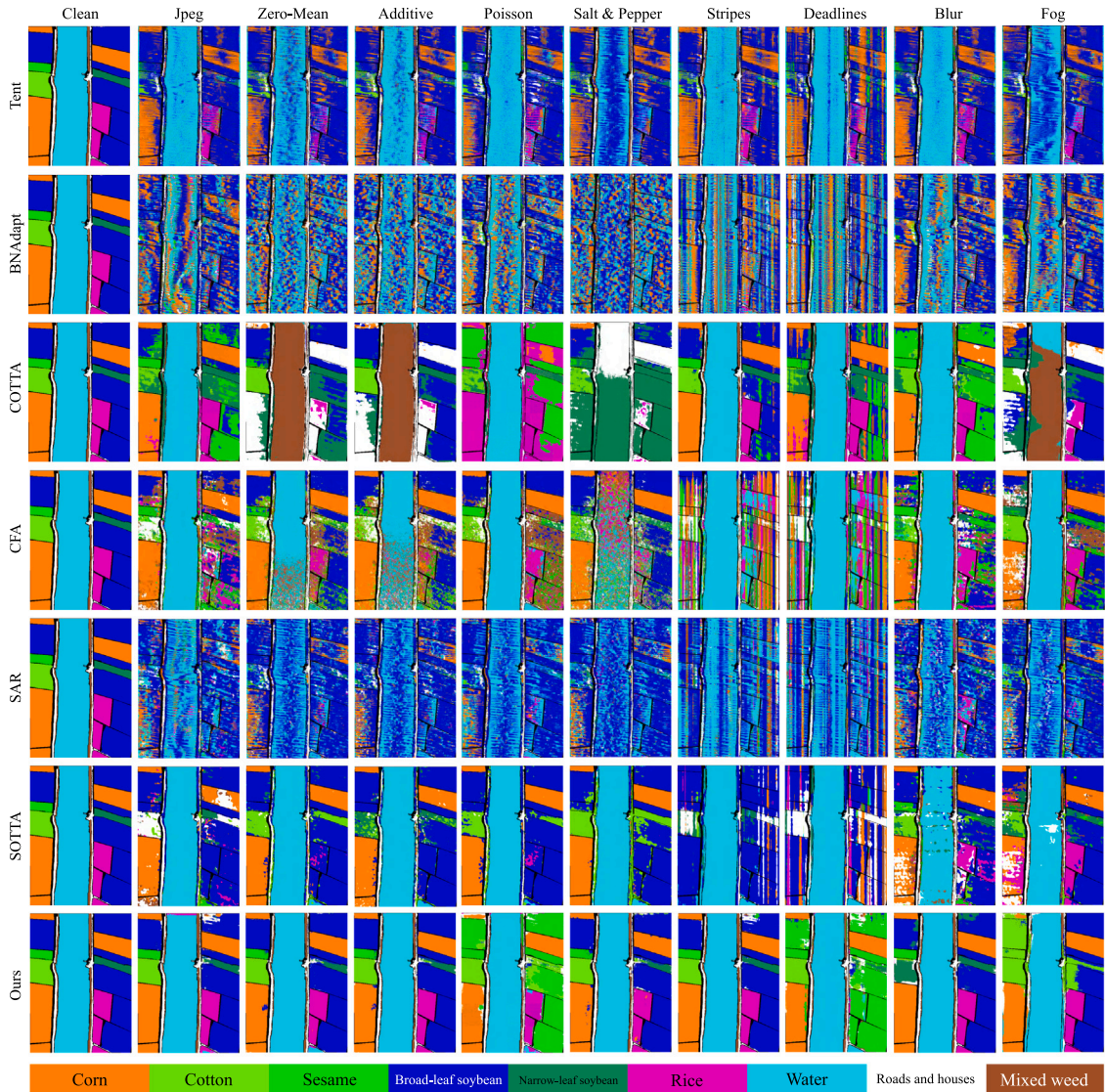


Fig. 9. Visual classification results on the WHLK dataset under different degradation types after applying different TTA method. Each row corresponds to a method, and each column represents a specific degradation.

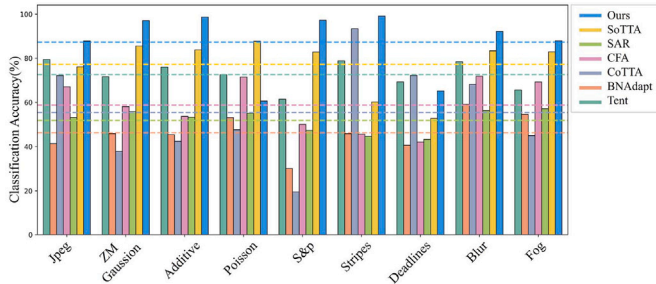


Fig. 10. Classification performance (OA) of different TTA methods under diverse degradation types on the WHLK dataset. Each group of bars corresponds to a specific degradation type, and the dashed lines represent the average accuracy of each method across all degradations.

spatial continuity and suppress isolated misclassifications caused by degraded local cues. In addition, the CELA module refines predictions at test time by selectively adapting reliable samples and suppressing uncertain responses during online adaptation, thereby preventing the amplification of noisy predictions commonly observed in entropy-based methods such as Tent.

Consequently, the combination of SSTC and CELA leads to smoother and more stable classification maps under severe degradation conditions. In contrast, the competing methods tend to suffer from spatial fragmentation, class confusion, and boundary distortion. Overall, these results demonstrate that HyperTTA consistently achieves superior robustness and performance stability across different degradation types, highlighting its practical potential for real-world HSI classification.

4.4. Ablation study

Table 5 presents the ablation study results on the three key components of the proposed framework, including the Multi-level Receptive Field (MRF) module, the Label Smoothing Loss (LSL) module, and the Confidence-aware Entropy-minimized LayerNorm Adapter (CELA) module. Each component is ablated individually to assess its contribution under various degradation types. This experiment is conducted on the WHLK dataset to ensure a comprehensive evaluation under complex and realistic conditions.

The complete model (“Ours”) consistently achieves the highest classification accuracy across most degradation types. Notably, it obtains 87.72% OA on JPEG noise, 97.08% on zero-mean Gaussian noise, and 87.86% on fog degradation, indicating strong robustness to diverse noise patterns. Removing the MRF module causes a significant performance drop under complex degradations such as fog (from 87.86% to 29.52%) and zero-mean Gaussian noise (from 97.08% to 52.07%), suggesting its

importance for modeling multi-scale spatial consistency. Excluding the LSL results in a moderate but consistent decline across most conditions, especially under blur and fog. The absence of the CELA module leads to clear degradation in performance under several noise types, such as blur (OA drops from 92.16% to 85.96%) and fog (from 87.86% to 64.09%).

Overall, these results demonstrate that each component contributes to the overall robustness of the proposed framework, and their combination leads to significantly improved performance across diverse degradation scenarios.

5. Conclusion

In this paper, we investigate the robustness of HSI classification under a wide range of degradations and propose a framework named HyperTTA (Test-Time Adaptable Transformer for Hyperspectral Degradation) to enhance model robustness against diverse degradation conditions. Specifically, we first construct a comprehensive multi-degradation hyperspectral dataset that simulates nine representative types of corruptions, enabling systematic evaluation of model performance under challenging scenarios. Based on this dataset, we develop a spectral-spatial Transformer-based classification framework that incorporates a multi-level receptive field mechanism and label smoothing regularization to jointly enhance spatial representation and generalization capability. Furthermore, we introduce a lightweight TTA strategy that updates only the affine parameters of LayerNorm layers via entropy minimization. To improve the stability and reliability of the adaptation process, this update is performed selectively on high-confidence unlabeled target samples. This confidence-aware strategy enables robust, source-free adaptation to distribution shifts without requiring access to source data or target annotations. Experimental results on PU and WHLK datasets confirm that HyperTTA achieves superior performance under various degradation conditions, demonstrating its robustness and generalization capability.

While HyperTTA demonstrates promising performance, it has several limitations. The current TTA strategy relies on clean source model weights, which may limit its applicability to models under extreme distribution shifts. Additionally, the degradation simulations are based on synthetic corruptions and do not yet include real-world degraded hyperspectral data. In future work, we will enhance the realism of the dataset by incorporating physically acquired degradations (e.g., fog, sensor failures), and extend the framework to dynamic scenarios such as temporally evolving scenes and online adaptation in streaming settings.

CRedit authorship contribution statement

Xia Yue: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.
Anfeng Liu: Writing – review & editing, Supervision, Resources, Data

Table 5
Ablation study on the MRF, LSL, and CELA modules on the WHLK dataset.

Method	Metric	Clean	Jpeg	Zero-Mean	Additive	Poisson	Salt & Pepper	Stripe	Deadline	Blur	Fog
Ours	OA (%)	99.80	87.72	97.08	98.70	60.58	97.25	99.10	65.16	92.16	87.86
	AA (%)	99.17	93.29	90.01	95.81	74.44	90.38	98.36	76.66	85.30	64.69
	Kappa × 100	99.73	85.32	96.15	98.29	54.11	96.38	98.82	58.98	89.83	83.27
w/o MRF	OA (%)	99.85	63.84	52.07	60.72	59.15	64.00	98.80	66.47	84.59	29.52
	AA (%)	99.57	75.95	58.67	66.96	78.80	68.94	99.03	81.37	80.03	35.48
	Kappa × 100	99.80	54.94	46.24	54.31	52.72	58.68	98.43	60.54	80.71	22.82
w/o LSL	OA (%)	99.66	69.82	71.97	93.93	61.68	78.21	96.44	64.58	86.45	54.40
	AA (%)	98.98	77.72	73.68	92.89	74.98	77.23	96.95	77.14	80.34	49.79
	Kappa × 100	99.55	62.27	66.39	91.94	55.34	73.56	95.41	58.39	82.77	46.58
w/o CELA	OA (%)	99.80	85.06	86.25	92.14	56.32	87.76	95.62	69.09	85.96	64.09
	AA (%)	99.17	92.11	84.07	90.20	70.43	89.08	96.83	80.65	81.21	63.18
	Kappa × 100	99.73	81.24	82.77	89.90	49.47	84.50	94.33	63.14	82.21	57.62

curation, Conceptualization. **Ning Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Chenjia Huang:** Writing – review & editing, Visualization, Validation. **Hui Liu:** Visualization, Validation, Investigation, Funding acquisition, Formal analysis. **Zhou Huang:** Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Leyuan Fang:** Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Author biography



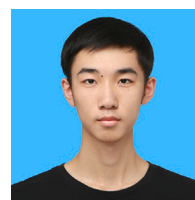
Xia Yue received a Bachelor's degree in Computer Science from the Central South University, Changsha, China, in 2020. She is currently pursuing the Ph.D. degree at the School of Computer Science and Engineering, Central South University, Changsha, China. Her research interests include machine learning, remote sensing, and HSI processing.



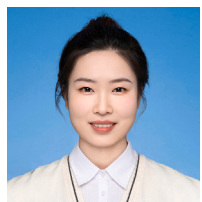
Anfeng Liu received the M.Sc. and Ph.D. degrees in computer science from Central South University, China, in 2002 and 2005, respectively. He is currently a Professor with the School of Electronic Information, Central South University. He has published four books and over 100 international journal and conference papers, among which there are more than 30 ESI highly-cited papers. His major research interests include wireless sensor networks, the Internet of Things, information security, edge computing, and crowdsourcing.



Ning Chen received the B.S. degree from the School of Earth and Space Sciences, Peking University, Beijing, China, in 2016, and the M.S. degree in GIS from the School of Earth and Space Sciences, Peking University, Beijing, China in 2019. He is currently an Engineer with the Institute of Remote Sensing and Geographic Information System, Peking University. His research interests include satellite image understanding, recommendation system, large-scale sparse learning and pattern recognition. He serves as a reviewer for IEEE Transactions on Image Processing, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Geoscience and Remote Sensing, IEEE Geoscience and Remote Sensing Letters, etc.



Chenjia Huang received the B.S. degree from Nanjing University of Information Science and Technology, Nanjing, China, in 2023. He is currently pursuing the M.S. degree in Surveying and Mapping Science and technology, Nanjing University of Information Science and Technology, Nanjing, China. His research interests include hyperspectral image classification and deep learning.



Hui Liu received a B.S. degree from the School of Mathematics and Statistics, Xidian University, Xi'an, China, in 2014, and a Ph.D. degree from the Institute of Remote Sensing and Geographical Information Systems, School of Earth and Space Sciences, Peking University, Beijing, China, in 2020. She was a visiting scholar with the Department of Geography and geoinformation science at George Mason University from 2018 to 2019, and a Post-Doctoral Research Fellow with the Institute of Remote Sensing and Geographical Information Systems, at Peking University from 2020 to 2022. She is now an associate professor and master tutor with the

Nanjing University of Information Science and Technology. She is focusing on research about spatiotemporal modeling, GIScience, deep learning and its application in remote sensing areas.



Zhou Huang received the B.S. degree in GIS and the Ph.D. degree in Cartography and GIS from Peking University, China, in 2004 and 2009, respectively. He is a tenured associate professor of GIScience at the Institute of Remote Sensing and Geographic Information System, Peking University. Currently, his main research interests include big geo-data, high-performance geocomputation, distributed geographic information processing, spatial data mining, and spatial database. He has published more than 100 academic papers in international journals or conferences. In addition, Zhou Huang serves as Deputy Director of Institute of Remote

Sensing and GIS, Peking University, and Deputy Director of Engineering Research Center of Earth Observation and Navigation, Ministry of Education, China. He was also selected for the Youth Talent Innovation Plan in Remote Sensing Science and Technology, funded by the Ministry of Science and Technology of China.



Leyuan Fang received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015. From August 2016 to September 2017, he was a Postdoc Researcher with the Department of Biomedical Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multi-resolution analysis in remote sensing and medical image processing. He is the associate editors of IEEE Transactions on Image Processing, IEEE Transactions on Geoscience and

Remote Sensing, IEEE Transactions on Neural Networks and Learning Systems, and Neurocomputing. He was a recipient of one 2nd- Grade National Award at the Nature and Science Progress of China in 2019.